EP2 – Exercise 4
One spherical collapse, three numerical formulations

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1 Model and equations

We follow the smooth-dark-energy, shear-free limit of Pace, Meyer & Bartelmann [1], as fixed in the Lecture 8 notes [2]. The background is flat ΛCDM (EdS for $\Omega_{m0} = 1$) with radiation neglected at all epochs:

$$E^2(a) = \Omega_{m0} a^{-3} + 1 - \Omega_{m0}, \qquad \Omega_m(a) = \frac{\Omega_{m0} a^{-3}}{E^2}, \qquad A_H = 2 + \frac{d\ln H}{dN} = 2 - \frac{3}{2}\Omega_m(a), \tag{1}$$

with $N = \ln a$, $X_N = dX/dN$, $\vartheta = \nabla_x \cdot v/(aH)$ and $g = 1/\delta$ (inverse overdensity, not the growth rate). The three nonlinear systems and the linear equation are

$$\text{A:} \quad \delta_{NN} + A_H \delta_N - \frac{4}{3}\frac{\delta_N^2}{1+\delta} - \frac{3}{2}\Omega_m \delta(1+\delta) = 0, \tag{2}$$

$$\text{B:} \quad \delta_N = -(1+\delta)\vartheta, \qquad \vartheta_N = -A_H\vartheta - \frac{\vartheta^2}{3} - \frac{3}{2}\Omega_m\delta, \tag{3}$$

$$\text{C:} \quad g_N = g(1+g)\vartheta, \qquad \vartheta_N = -A_H\vartheta - \frac{\vartheta^2}{3} - \frac{3\Omega_m}{2g}, \tag{4}$$

$$\text{linear:} \quad \delta_{L,NN} + A_H \delta_{L,N} - \frac{3}{2}\Omega_m \delta_L = 0. \tag{5}$$

A is integrated as the first-order pair (δ, δ_N), B as (δ, ϑ) and C as (g, ϑ), each with its own right-hand side and its own call to the integrator. Algebraic equivalence was checked at 10^3 random states ($10^{-5} < \delta < 10^8$): differentiating B's continuity equation reproduces A's δ_{NN}, and C at $g = 1/\delta$ reproduces $g_N = -\delta_N/\delta^2$ and B's ϑ_N, to $\lesssim 10^{-14}$. The three independent trajectories agree to $\lesssim 10^{-9}$ until just before collapse (Fig. 1).

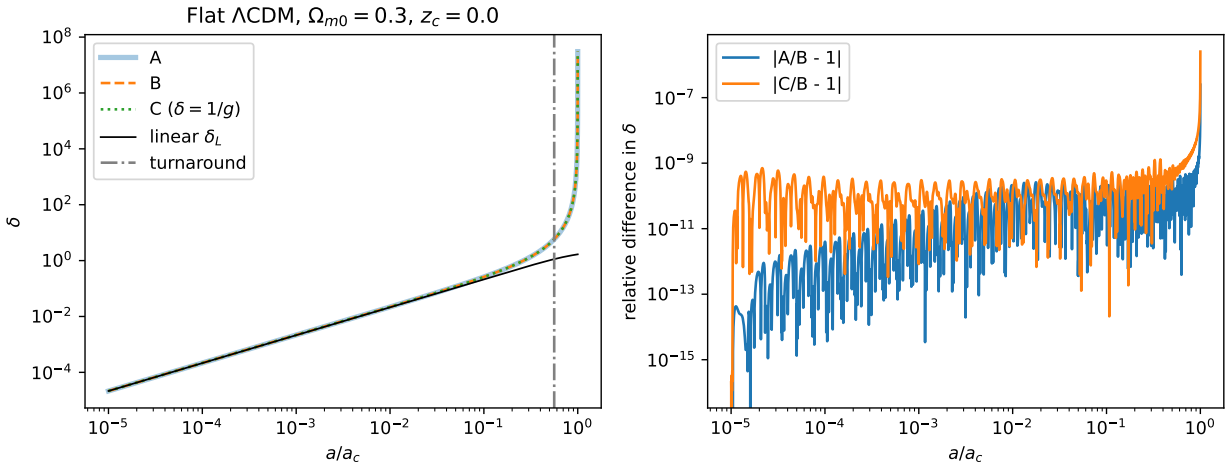


Figure 1: Left: $\delta(a)$ from A, B, C (same δ_i) and the linear δ_L; flat ΛCDM, $\Omega_{m0} = 0.3$, $z_c = 0$. Right: relative differences between the independent integrations.

2 Initial conditions, events and shooting

For a trial δ_i at $a_i \ll a_c$ (growing mode, notes eq. 6.6):

$$\delta_{N,i} = f_i \delta_i, \qquad \vartheta_i = -\frac{f_i \delta_i}{1+\delta_i}, \qquad g_i = \delta_i^{-1}, \qquad \delta_{L,i} = \delta_i, \quad \delta_{L,N,i} = f_i \delta_i, \tag{6}$$

where $f_i = d\ln D / d\ln a$ at a_i is taken from the Exercise 1 growth ODE started at $a_i/100$ ($1 - f_i \lesssim 10^{-9}$ for $a_i \le 10^{-3}$).

- **Collapse event** (terminal): $\delta = \delta_\infty$ for A and B, $g = \epsilon$ for C. The integration stops there and never crosses the singular endpoint.
- **Turnaround event**: $d\ln R/dN = 1 + \vartheta/3 = 0$, i.e. $\vartheta = -3$ (A uses $\vartheta = -\delta_N/(1+\delta)$, C uses $\delta = 1/g$); $\zeta_{\rm ta} = 1 + \delta(a_{\rm ta})$.
- **Shooting**: Brent's method on $F(\delta_i) = N_{\rm event}(\delta_i) - \ln a_c$. The bracket starts at $[0.9, 1.1] \times 1.686\, D(a_i)/D(a_c)$ and is widened until F changes sign. Relative tolerance on δ_i: 10^{-13}.
- $\delta_c = \delta_L(a_c)$ from the linear solution with the tuned δ_i.
- **Defaults**: $a_i = 10^{-5}$, `scipy.integrate.solve_ivp` with `DOP853`, `rtol`= 10^{-10}, `atol`= 10^{-14}, $\delta_\infty = 10^8$, $\epsilon = 10^{-8}$.

Table 1 lists the required outputs for $z_c = 0$, including brackets, number of iterations and event residuals.

Table 1: Required outputs for $z_c = 0$ (default settings). $\ln(a_{\rm ev}/a_c)$ is the event residual.

Ω_{m0}	form	δ_i	bracket lo	bracket hi	iter.	$\ln(a_{\rm ev}/a_c)$	$a_{\rm ta}$	$\zeta_{\rm ta}$	δ_c	$\Delta_{{\rm vir},m}$	$\Delta_{{\rm vir},c}$
1.0	A	1.6863898036e-05	1.5174e-05	1.8546e-05	7	-5.9e-16	0.630003	5.551652	1.686390	177.617	177.617
1.0	B	1.6863898036e-05	1.5174e-05	1.8546e-05	7	-5.4e-17	0.630003	5.551652	1.686390	177.617	177.617
1.0	C	1.6863898034e-05	1.5174e-05	1.8546e-05	6	1.3e-15	0.630003	5.551652	1.686390	177.617	177.617
0.3	A	2.1508359780e-05	1.9479e-05	2.3808e-05	7	-4.0e-15	0.561462	6.837308	1.675460	339.714	101.914
0.3	B	2.1508359781e-05	1.9479e-05	2.3808e-05	7	-1.6e-16	0.561462	6.837308	1.675460	339.714	101.914
0.3	C	2.1508359778e-05	1.9479e-05	2.3808e-05	7	-6.4e-16	0.561462	6.837308	1.675460	339.714	101.914

3 Virial prescription

The same prescription is applied to the outputs of A, B and C (notes Section 3.3). With $y = R_{\rm vir}/R_{\rm ta}$ and

$$\lambda_{\rm ta} = \frac{\rho_\Lambda}{\rho_{\rm sph,ta}} = \frac{\Omega_\Lambda(a_{\rm ta})}{\Omega_m(a_{\rm ta})\,\zeta_{\rm ta}}, \qquad 4\lambda_{\rm ta} y^3 - 2(1+\lambda_{\rm ta}) y + 1 = 0, \tag{7}$$

we take the root in $(0, 1/2]$: the cubic equals $+1$ at $y = 0$ and $-\lambda_{\rm ta}/2$ at $y = 1/2$, and this root tends continuously to $y = 1/2$ as $\lambda_{\rm ta} \to 0$ (EdS). Using the formal-collapse epoch,

$$\Delta_{{\rm vir},m}(a_c) = \zeta_{\rm ta} \left(\frac{a_c}{a_{\rm ta}}\right)^3 y^{-3}, \qquad \Delta_{{\rm vir},c}(a_c) = \Omega_m(a_c)\, \Delta_{{\rm vir},m}(a_c). \tag{8}$$

4 EdS benchmarks

Table 2 compares with $\delta_c = \frac{3}{20}(12\pi)^{2/3} = 1.68647$, $a_{\rm ta}/a_c = 2^{-2/3}$, $\zeta_{\rm ta} = (3\pi/4)^2$, $\Delta_{{\rm vir},m} = 18\pi^2$, at four collapse redshifts. $\zeta_{\rm ta}$ is recovered to $\sim 10^{-10}$. $a_{\rm ta}/a_c$ ($+6.7 \times 10^{-5}$) and $\Delta_{{\rm vir},m}$ (-2.0×10^{-4}) have the same error at every z_c: it is the finite-threshold bias, since the event fires slightly before the true collapse. δ_c has, in addition to this bias ($\approx -6.7 \times 10^{-5}$), the initial-epoch error $\propto \delta_i \propto a_i/a_c$

Table 2: EdS validation: relative errors with respect to the exact values (default settings).

z_c	form	δ_c	err δ_c	err $a_{\rm ta}/a_c$	err $\zeta_{\rm ta}$	err $\Delta_{{\rm vir},m}$	$\ln(a_{\rm ev}/a_c)$
0	A	1.686390	-4.77e-05	+6.68e-05	+2.56e-11	-2.00e-04	-5.9e-16
0	B	1.686390	-4.77e-05	+6.68e-05	+2.08e-12	-2.00e-04	-5.4e-17
0	C	1.686390	-4.77e-05	+6.68e-05	-5.28e-11	-2.00e-04	1.3e-15
1	A	1.686422	-2.86e-05	+6.68e-05	-1.11e-11	-2.00e-04	1.1e-16
1	B	1.686422	-2.86e-05	+6.68e-05	+8.52e-12	-2.00e-04	2.7e-15
1	C	1.686422	-2.86e-05	+6.68e-05	-4.91e-11	-2.00e-04	-3.3e-16
2	A	1.686454	-9.45e-06	+6.68e-05	-1.06e-11	-2.00e-04	-4.4e-16
2	B	1.686454	-9.45e-06	+6.68e-05	+8.44e-12	-2.00e-04	0.0e+00
2	C	1.686454	-9.45e-06	+6.68e-05	-5.80e-11	-2.00e-04	2.2e-16
5	A	1.686551	+4.79e-05	+6.68e-05	-4.41e-11	-2.00e-04	-2.2e-15
5	B	1.686551	+4.79e-05	+6.68e-05	+1.85e-12	-2.00e-04	-1.3e-15
5	C	1.686551	+4.79e-05	+6.68e-05	-9.24e-11	-2.00e-04	0.0e+00

(Section 6), which explains its small drift with z_c at fixed a_i. No threshold was tuned to match any single benchmark.

The finite-threshold bias scales as $\delta_\infty^{-1/2}$ (Fig. 2). This follows from the cycloid (notes eq. 2.6): near $\eta = 2\pi$, $R \propto (t_c - t)^{2/3}$, so $\delta \propto (t_c - t)^{-2}$ and the event at δ_∞ comes early by $\Delta t/t_c \propto \delta_\infty^{-1/2}$. With $\epsilon = 1/\delta_\infty$, A, B and C give identical errors: the bias belongs to the event, not to the formulation.

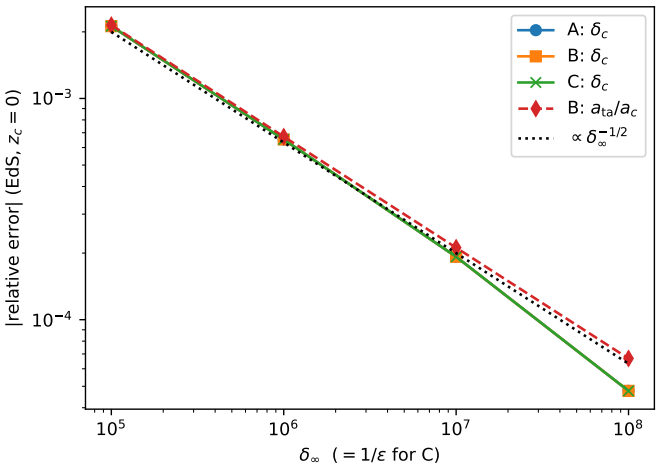


Figure 2: EdS, $z_c = 0$: relative error in δ_c (A, B, C) and $a_{\rm ta}/a_c$ (B) versus the event threshold.

5 Flat ΛCDM grid

Figure 3 and Table 3 show $\Omega_{m0} \in \{0.2, 0.3, 0.4\}$, $\Omega_{\Lambda 0} = 1 - \Omega_{m0}$, $z_c \in \{0, 1, 2\}$ (curves use a finer z_c step of 0.25). Over the full grid the spread between A, B and C is $\leq 2 \times 10^{-10}$ in δ_c, $\zeta_{\rm ta}$ and $\Delta_{{\rm vir},m}$.

Roles of the different ingredients (Table 4).

- **Growth.** Λ slows the linear growth and the nonlinear collapse by almost the same amount, so δ_c decreases by less than 1% (lowest Ω_{m0}, $z_c = 0$: $0.991\,\delta_c^{\rm EdS}$).

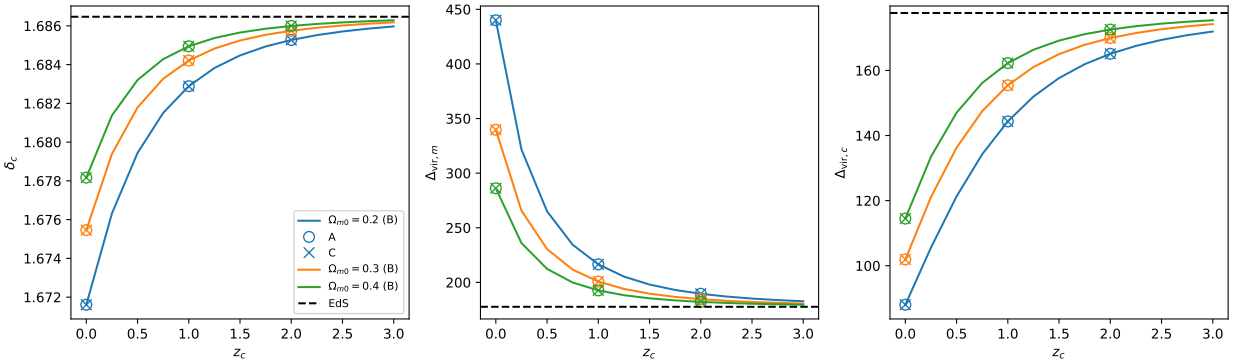


Figure 3: δ_c, $\Delta_{\mathrm{vir},m}$ and $\Delta_{\mathrm{vir},c}$ versus z_c. Lines: B; open markers: A and C at $z_c = 0, 1, 2$; dashed: EdS.

Table 3: Flat ΛCDM grid (formulation B; A and C agree to $\leq 2 \times 10^{-10}$).

Ω_{m0}	z_c	a_{ta}/a_c	ζ_{ta}	y	δ_c	$\Delta_{\mathrm{vir},m}$	$\Delta_{\mathrm{vir},c}$
0.2	0	0.536658	7.44927	0.478472	1.671608	439.997	87.999
0.2	1	0.608035	5.91029	0.495201	1.682886	216.509	144.339
0.2	2	0.622669	5.66661	0.498417	1.685272	189.571	165.110
0.3	0	0.561462	6.83731	0.484474	1.675460	339.714	101.914
0.3	1	0.616291	5.77039	0.497025	1.684212	200.777	155.440
0.3	2	0.625622	5.61977	0.499057	1.685748	184.643	169.955
0.4	0	0.578660	6.46497	0.488520	1.678169	286.185	114.474
0.4	1	0.620856	5.69575	0.498023	1.684947	192.678	162.255
0.4	2	0.627152	5.59579	0.499387	1.685994	182.151	172.564

- **Turnaround.** The sphere turns around denser (ζ_{ta} up to 1.34× EdS) and earlier relative to collapse ($a_{\mathrm{ta}}/a_c < 2^{-2/3}$), so the background dilutes more until a_c (factor up to 1.62). These are the two largest factors in $\Delta_{\mathrm{vir},m}$.
- **Virial prescription.** The Λ potential energy gives $y < 1/2$: a smaller, denser virialized halo (factor up to 1.14).
- **Reference density.** $\Delta_{\mathrm{vir},m}$ rises above $18\pi^2$ while $\Delta_{\mathrm{vir},c} = \Omega_m(a_c)\Delta_{\mathrm{vir},m}$ falls below it, because $\Omega_m(a_c) < 1$: the same halo shows opposite trends for the two conventions.

At high z_c matter dominates and all quantities approach the EdS lines.

6 Convergence and failure modes

Flat ΛCDM, $\Omega_{m0} = 0.3$, $z_c = 0$. One setting is changed at a time and δ_i is retuned in every run. The reference is B with `rtol`$= 10^{-12}$, `atol`$= 10^{-16}$ ($\delta_c = 1.6754603968$, $\Delta_{\mathrm{vir},m} = 339.714231$, $\Delta_{\mathrm{vir},c} = 101.914269$); it shares a_i and the threshold with the defaults, so those rows measure differences with respect to the default, not to the exact answer.

- **All formulations.** (i) Threshold bias $\propto \delta_\infty^{-1/2}$; from this scaling the error left at $\delta_\infty = 10^8$ is $\approx 6 \times 10^{-5}$ in δ_c and $\approx 2 \times 10^{-4}$ in Δ_{vir}, as in EdS. It is the dominant error at the default settings. (ii) Initial epoch: eq. (6.6) is the growing mode only to first order in δ_i, so the δ_c error scales as $\delta_i \propto a_i$ (2.4×10^{-3} at $a_i = 10^{-3}$, 2.2×10^{-4} at 10^{-4}). ζ_{ta} and Δ_{vir} do not move with a_i: the shooting absorbs the initial error in the nonlinear run, while δ_c inherits it through δ_L. Event residuals stay $\lesssim 10^{-13}$, so the root search is never the limitation.

Table 4: Factor decomposition: $\Delta_{\mathrm{vir},m}/18\pi^2 = [\zeta_{\mathrm{ta}}/\zeta_{\mathrm{ta}}^{\mathrm{EdS}}] \times [(a_c/a_{\mathrm{ta}})^3/4] \times (2y)^{-3}$, and $\Delta_{\mathrm{vir},c}/\Delta_{\mathrm{vir},m} = \Omega_m(a_c)$.

Ω_{m0}	z_c	$\delta_c/\delta_c^{\mathrm{EdS}}$	$\zeta_{\mathrm{ta}}/\zeta_{\mathrm{ta}}^{\mathrm{EdS}}$	$(a_c/a_{\mathrm{ta}})^3/4$	$(2y)^{-3}$	$\Delta_{\mathrm{vir},m}/18\pi^2$	$\Omega_m(a_c)$
0.2	0	0.9912	1.3418	1.6175	1.1411	2.4767	0.2000
0.2	1	0.9979	1.0646	1.1121	1.0294	1.2187	0.6667
0.2	2	0.9993	1.0207	1.0355	1.0096	1.0671	0.8710
0.3	0	0.9935	1.2316	1.4125	1.0993	1.9122	0.3000
0.3	1	0.9987	1.0394	1.0680	1.0181	1.1302	0.7742
0.3	2	0.9996	1.0123	1.0209	1.0057	1.0393	0.9205
0.4	0	0.9951	1.1645	1.2902	1.0722	1.6109	0.4000
0.4	1	0.9991	1.0260	1.0446	1.0120	1.0846	0.8421
0.4	2	0.9997	1.0080	1.0135	1.0037	1.0253	0.9474

- **A.** δ and δ_N both diverge and the source contains $\delta_N^2/(1+\delta)$. Early on $\delta \sim 10^{-5}$, so `atol` must be $\ll \delta_i$; `atol`$= 10^{-6}$ gives 6×10^{-5} in δ_c.
- **B.** Same `atol`–δ_i sensitivity as A (5×10^{-5}); δ and $|\vartheta|$ diverge at collapse.
- **C.** g runs from $\sim 5 \times 10^4$ down to ϵ, so `atol` must be $\ll \epsilon$, otherwise the event $g = \epsilon$ is not resolved: `atol`$= 10^{-6}$ gives 2×10^{-3}, the largest error of all. C is also the most sensitive to a loose `rtol` and a lower-order integrator (10^{-5} with RK45), as the $1/g$ source and ϑ still diverge.

With tight settings A, B and C agree to $\sim 10^{-10}$, far below the threshold and a_i effects. We do not reproduce the statement that C is less noisy [1]: with adaptive step control and event location, C is as precise as A and B at tight settings and less robust at loose ones.

References

[1] F. Pace, S. Meyer and M. Bartelmann, "On the implementation of the spherical collapse model for dark energy models," JCAP **10**, 040 (2017), arXiv:1708.02477.

[2] T. Castro, *Lecture 8 Notes: Spherical Collapse and Halo Abundances*, Numerical Cosmology, IFUSP; Sections 2, 3, 6 and 7.

[3] P. Virtanen et al., "SciPy 1.0: fundamental algorithms for scientific computing in Python," Nature Methods **17**, 261 (2020).

Table 5: Relative change of δ_c with respect to the reference, cross-formulation difference and worst event residual. Threshold setting: δ_∞ / ϵ; tolerances: `rtol` / `atol`; "loose atol" uses `rtol`$= 10^{-10}$; "method" uses `rtol`$= 10^{-6}$.

test	setting	$\Delta\delta_c/\delta_c$ A	$\Delta\delta_c/\delta_c$ B	$\Delta\delta_c/\delta_c$ C	max cross-form.	max $\lvert\ln(a_{\rm ev}/a_c)\rvert$
initial epoch	1e-03	+2.4e-03	+2.4e-03	+2.4e-03	+6.3e-11	+3.9e-15
initial epoch	1e-04	+2.2e-04	+2.2e-04	+2.2e-04	+8.7e-11	+4.5e-15
initial epoch	1e-05	+4.1e-11	+5.6e-11	-9.2e-11	+1.5e-10	+4.0e-15
threshold	1e+05 / 1e-05	-2.0e-03	-2.0e-03	-2.0e-03	+1.5e-10	+6.6e-15
threshold	1e+06 / 1e-06	-5.8e-04	-5.8e-04	-5.8e-04	+1.5e-10	+8.3e-15
threshold	1e+07 / 1e-07	-1.4e-04	-1.4e-04	-1.4e-04	+1.5e-10	+1.9e-15
threshold	1e+08 / 1e-08	+4.1e-11	+5.6e-11	-9.2e-11	+1.5e-10	+4.0e-15
tolerances	1e-06 / 1e-10	+5.4e-07	+9.0e-07	-2.4e-06	+3.3e-06	+1.3e-14
tolerances	1e-08 / 1e-12	+5.6e-09	+6.9e-09	-1.2e-08	+1.9e-08	+7.5e-15
tolerances	1e-10 / 1e-14	+4.1e-11	+5.6e-11	-9.2e-11	+1.5e-10	+4.0e-15
tolerances	1e-12 / 1e-16	-9.8e-14	+0.0e+00	-1.1e-12	+1.1e-12	+2.8e-15
loose atol	1e-06	-5.9e-05	-5.5e-05	-2.1e-03	+2.0e-03	+2.5e-13
loose atol	1e-08	-4.8e-07	-1.1e-07	-3.4e-05	+3.4e-05	+3.9e-15
loose atol	1e-10	-6.0e-09	-2.4e-09	-3.8e-07	+3.8e-07	+3.5e-15
method	RK45	+7.4e-07	+6.6e-07	+1.0e-05	+9.7e-06	+8.0e-16
method	DOP853	+6.6e-07	+6.2e-07	-1.5e-06	+2.1e-06	+1.4e-14

Table 6: Same as Table 5 for $\Delta_{{\rm vir},m}$.

test	setting	$\Delta\Delta_{{\rm vir},m}/\Delta_{{\rm vir},m}$ A	$\Delta\Delta_{{\rm vir},m}/\Delta_{{\rm vir},m}$ B	$\Delta\Delta_{{\rm vir},m}/\Delta_{{\rm vir},m}$ C
initial epoch	1e-03	-2.3e-11	-2.8e-11	-2.3e-11
initial epoch	1e-04	+5.8e-11	+1.1e-10	+2.8e-11
initial epoch	1e-05	+8.3e-11	+7.8e-11	-3.9e-11
threshold	1e+05 / 1e-05	-6.5e-03	-6.5e-03	-6.5e-03
threshold	1e+06 / 1e-06	-1.9e-03	-1.9e-03	-1.9e-03
threshold	1e+07 / 1e-07	-4.5e-04	-4.5e-04	-4.5e-04
threshold	1e+08 / 1e-08	+8.3e-11	+7.8e-11	-3.9e-11
tolerances	1e-06 / 1e-10	+1.4e-06	+2.2e-07	-2.0e-07
tolerances	1e-08 / 1e-12	+2.4e-09	+4.8e-09	-8.9e-09
tolerances	1e-10 / 1e-14	+8.3e-11	+7.8e-11	-3.9e-11
tolerances	1e-12 / 1e-16	-2.7e-13	+0.0e+00	+3.2e-14
loose atol	1e-06	+1.8e-08	+7.3e-08	-1.9e-07
loose atol	1e-08	-3.4e-10	-1.5e-09	-2.2e-09
loose atol	1e-10	-8.9e-11	+6.3e-11	+1.3e-11
method	RK45	+5.6e-07	+6.7e-07	+5.0e-07
method	DOP853	-9.7e-07	+1.1e-07	-3.6e-07